

THE EFFECTS OF FIBRE ORIENTATION ON FREE VIBRATIONS OF COMPOSITE BEAMS

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The torsion–flexure coupling effect of generally orthotropic beams is dependent on reinforcing fibre orientation and mode order. At higher ranks of vibration, this coupling effect is principally contributed by the twisting moment induced by bending. The influence of fibre orientation on normal mode shapes is more significant for small values of fibre orientation. The mode shapes can change suddenly with a small increase in fibre orientation.

1. INTRODUCTION

Fibre reinforced composite materials have been widely used in applications ranging from laminated aircraft wings in the aeronautical industry to brake pedal assemblies in the automobile industry. The increasing usage of composite materials in preference to conventional materials in structural applications is attributed to their high strength-to-density ratios and other properties, such as resistance to fatigue and low coefficient of friction. Generally orthotropic materials consist of much stiffer, continuous, unidirectional reinforcing fibres embedded in a relatively soft matrix. The potential of this class of fibre reinforced materials in structural applications has stimulated a considerable amount of study in the field of mechanics of such materials.

Dudek [1] and Ritchie *et al.* [2] have determined the elastic constants by using the resonant beam method. Several mathematical models [3–6] have been formulated to investigate the free vibrational response of generally orthotropic beams. Generally, there are two groups of mathematical models: discrete and continuous. Within each of these groups, there are two subgroups: (i) models which include the secondary effects, namely the shear deformation and rotary inertia effects, and (ii) models which do not include the secondary effects. The emphasis in references [3–6] is on mathematical modelling of free vibrational response of generally orthotropic beams, and the accuracy in the prediction of their natural frequencies. Also reported in the references are the presence of the torsion–flexure coupling effect and the influence of fibre orientation on natural frequencies. However, to the authors' knowledge, a detailed discussion of the effects of the fibre orientation on mode shapes and of the torsion–flexure coupling effect has not been previously presented.

2. EQUATIONS OF MOTION

The formulation in reference [5] shows that Timoshenko's theory of flexure [7] and Lekhnitskii's theory of static torsion–flexure coupling [8] can be incorporated in Hamilton's principle. By carrying out the appropriate variational processes, the governing equations and boundary conditions of cantilevered generally orthotropic beams can be derived.

Thus, applying Hamilton's principle, one can write (a list of nomenclature is given in the Appendix)

$$\delta \int_{t=t_1}^{t=t_2} \int_0^L \left\{ \left[\frac{1}{2} \rho A \left(\frac{\partial v}{\partial t} \right)^2 + \frac{1}{2} \rho J_z \left(\frac{\partial \phi}{\partial t} \right)^2 + \frac{1}{2} \rho I_x \left(\frac{\partial \psi}{\partial t} \right)^2 \right] - \left[\frac{1}{2} M \frac{\partial \psi}{\partial z} + \frac{1}{2} T \frac{\partial \phi}{\partial z} + \frac{1}{2} \kappa A G_{zy} \left(\frac{\partial v}{\partial z} - \psi \right)^2 \right] \right\} dz dt = 0. \quad (1)$$

The expressions for the bending moment, M , and twisting moment, T , are

$$M = \frac{E_{zz} I_x}{\eta} \frac{\partial \psi}{\partial z} + \frac{C_t E_{zz} I_x}{\eta C_{mt}} \frac{\partial \phi}{\partial z}, \quad T = \frac{C_t \partial \phi}{\eta \partial z} + \frac{C_t E_{zz} I_x}{\eta C_{mt}} \frac{\partial \psi}{\partial z}. \quad (2, 3)$$

Carrying out the variational processes yields the governing differential equations,

$$\begin{aligned} \kappa A G_{zy} \frac{\partial}{\partial z} \left(\frac{\partial v}{\partial z} - \psi \right) &= \rho A \frac{\partial^2 v}{\partial t^2}, \\ \frac{E_{zz} I_x}{\eta} \frac{\partial^2 \psi}{\partial z^2} + \frac{C_t E_{zz} I_x}{\eta C_{mt}} \frac{\partial^2 \phi}{\partial z^2} + \kappa A G_{zy} \left(\frac{\partial v}{\partial z} - \psi \right) &= \rho I_x \frac{\partial^2 \psi}{\partial t^2}, \\ \frac{E_{zz} I_x}{\eta} \frac{C_t}{C_{mt}} \frac{\partial^2 \psi}{\partial z^2} + \frac{C_t}{\eta} \frac{\partial^2 \phi}{\partial z^2} &= \rho J_z \frac{\partial^2 \phi}{\partial t^2}, \end{aligned} \quad (4)$$

and the boundary conditions, which are that either member of each of the following three pairs vanishes: $(M\psi)$, $T(\phi)$, (Qv) .

3. DISCUSSION

The unidirectional graphite/epoxy composite beams described in reference [3] are adopted for the purpose of this discussion. The material parameters are as follows: $G_{12} = 2.54 \text{ GPa}$ ($0.3686 \times 10^6 \text{ psi}$), $E_{11}/G_{12} = 3.7$, $G_{23} = 5.16 \text{ GPa}$ ($0.7479 \times 10^6 \text{ psi}$), $E_{33}/G_{23} = 30.0$, $G_{13} = 5.16 \text{ GPa}$ ($0.7479 \times 10^6 \text{ psi}$), $\nu = 0.3$, $\rho = 48.18 \text{ kg/m}^3$ (0.056 lb/in^3). The beam dimensions are as follows: width = 30.48 mm, thickness = 3.175 mm, length = 190.5 mm. The boundary conditions for cantilevered beams are $v = 0$, $\phi = 0$ and $\psi = 0$ at the fixed end; at the free end, $M = 0$, $T = 0$ and $Q = 0$. The reference axes are defined as in Figure 1.

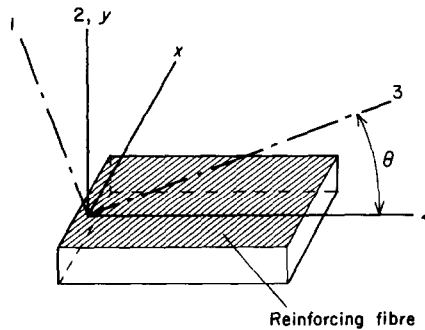


Figure 1. Cartesian co-ordinate systems for a generally orthotropic beam. The xy co-ordinate system denotes beam principal axes. The 1 2 3 co-ordinate system denotes material axes of elastic symmetry.

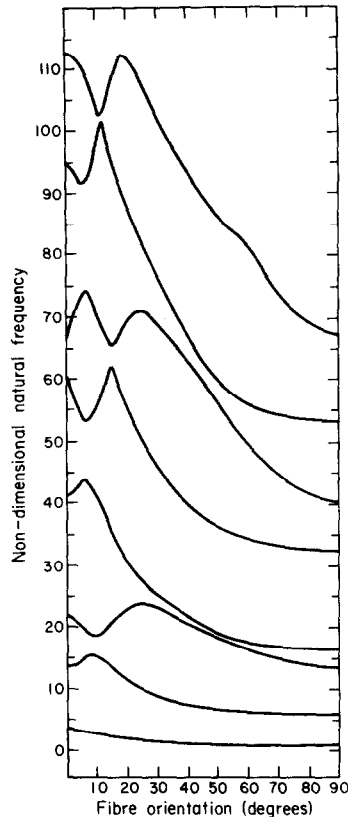


Figure 2. Variation of the first eight non-dimensional natural frequencies with fibre orientation.

By following the procedures described in reference [5], the characteristic equation can be obtained as

$$m^6 + \left(\frac{b^2 e^2 q^2}{\lambda} + b^2 e^2 r^2 + b^2 s^2 e^2 \right) m^4 + \left(\frac{b^4 e^4 q^2}{\lambda} (\eta r^2 + s^2) - b^2 e^2 (1 - b^2 s^2 e^2 r^2) \right) m^2 - \frac{\eta b^4 e^4 q^2}{\lambda} (1 - b^2 s^2 r^2 e^2) = 0. \quad (5)$$

The solutions are of the form

$$V_0(\xi) = A_1 \cosh(\sigma_1 \xi) + A_2 \sinh(\sigma_1 \xi) + A_3 \cos(\sigma_2 \xi) + A_4 \sin(\sigma_2 \xi) + A_5 \cos(\sigma_3 \xi) + A_6 \sin(\sigma_3 \xi),$$

$$\Psi_0(\xi) = C_1 [A_1 \sinh(\sigma_1 \xi) + A_2 \cosh(\sigma_1 \xi)] + C_2 [A_3 \sin(\sigma_2 \xi) - A_4 \cos(\sigma_2 \xi)] + C_3 [A_5 \sin(\sigma_3 \xi) - A_6 \cos(\sigma_3 \xi)],$$

$$\phi_0(\xi) = C_1 D_1 [A_1 \sinh(\sigma_1 \xi) + A_2 \cosh(\sigma_1 \xi)] + C_2 D_2 [A_3 \sin(\sigma_2 \xi) - A_4 \cos(\sigma_2 \xi)] + C_3 D_3 [A_5 \sin(\sigma_3 \xi) - A_6 \cos(\sigma_3 \xi)],$$

where

$$C_1 = (b^2 e^2 s^2 + \sigma_1^2) / \sigma_1, \quad C_2 = (b^2 e^2 s^2 - \sigma_2^2) / \sigma_2, \quad C_3 = (b^2 e^2 s^2 - \sigma_3^2) / \sigma_3, \\ D_1 = -(\alpha \sigma_1^2) / (b^2 e^2 q^2 \eta + \lambda \sigma_1^2), \quad D_2 = (\alpha \sigma_2^2) / (b^2 e^2 q^2 \eta - \lambda \sigma_2^2), \quad D_3 = (\alpha \sigma_3^2) / (b^2 e^2 q^2 \eta - \lambda \sigma_3^2),$$

σ_i ($i = 1, 2, 3$) are the solution roots of equation (5), and A_i ($i = 1, \dots, 6$) are undetermined amplitudes with the implication that only the mode shapes can be determined uniquely. The mode shapes which are referred to in later discussion have been normalized by letting $A_6 = 1$.

Figure 2 shows the variation of the first eight non-dimensional natural frequencies with reinforcing fibre orientations ranging from 0 to 90 degrees. Generally, the sudden increases and decreases in the curves are restricted to the region where fibre orientations are less than 25 degrees. One reason for this is that in this region the torsion-flexure coupling effect is most significant. The variation of the torsion-flexure coupling effect with reinforcing fibre orientation can be examined by investigating equations for bending and twisting moments. The second terms on the right-hand sides of equations (2) and (3) are responsible for the coupling between twisting and bending effects. Figure 3 shows the variation of the ratio

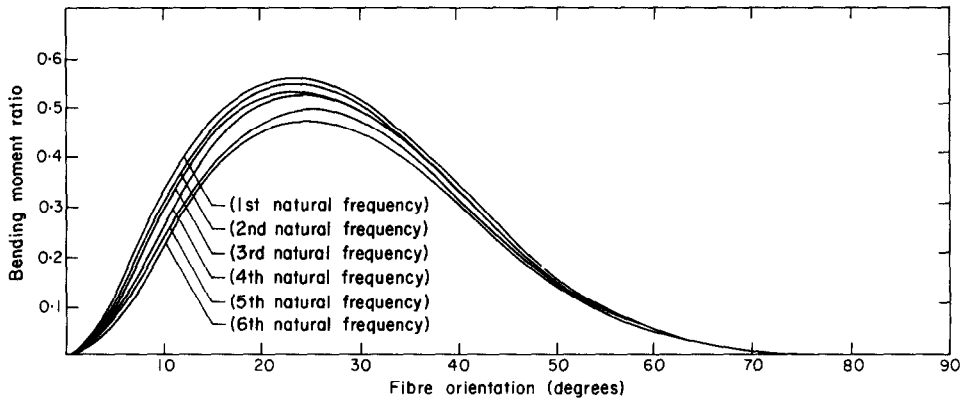


Figure 3. Variation of bending moment induced by twisting with fibre orientation.

formed by the second term to the first term in equation (2) with reinforcing fibre orientation, and hence this figure shows the variation, with fibre orientation, of the bending moment induced by twisting. Figure 4 shows the variation of the ratio formed by the second term to the first term in equation (3) with fibre orientation, and this figure shows the variation of twisting moment reduced by bending. Figure 3 shows that the magnitude of the bending moment induced by twisting as compared to that of a pure bending moment is always less than unity for the fibre orientations indicated; with the coupling effect decreasing with

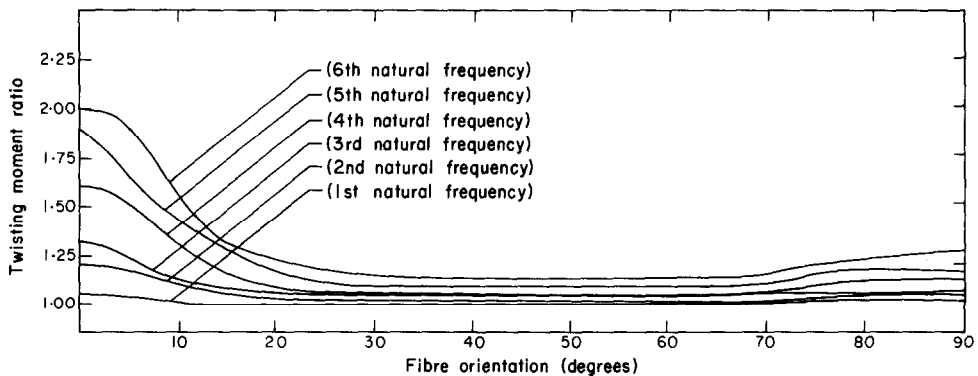


Figure 4. Variation of twisting moment induced by bending with fibre orientation.

increasing mode order. Figure 4 shows that the magnitude of the twisting moment is always greater than unity; with the coupling effect increasing with mode order. The effect of the bending moment induced by twisting reaches a maximum when the orientation of reinforcing fibres is 23 degrees. It should be noted that the maximum coupling effect does not always occur at 23 degrees. It varies from 24 to 25 degrees for the first five natural frequencies.

Before proceeding to consider the effect of fibre orientation on mode shapes, a comment on the terms used to describe the vibrational modes is in order. Abarcar and Cunniff [3] have described vibrational modes of generally orthotropic beams as either flexural or torsional. These two terms are generally used to describe vibrational modes of conventional materials. Using these terms for generally orthotropic beams suggests that vibrational modes are purely flexural or torsional, as in the case of beams of conventional materials. However, it has been shown in Figures 3 and 4 that for generally orthotropic beams the torsion-flexure coupling effect always exists and varies with fibre orientation. Ritchie *et al.* [2] described vibrational modes of generally orthotropic beams as either predominantly flexural or torsional. As these terms clearly suggest the presence of a coupling effect, they are adopted in this paper.

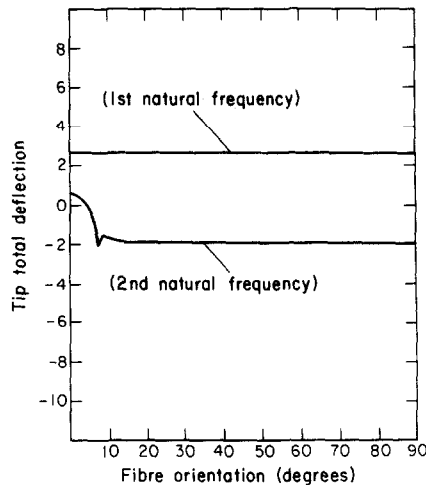


Figure 5. Variation of tip total deflection corresponding to first and second natural frequencies of a cantilevered graphite/epoxy beam with fibre orientation.

Consider now the variation of normal modes with fibre orientation. Instead of showing the variation of normal modes corresponding to small increment of fibre orientation for a specific order of natural frequency, the total deflection of the free end of the beam has been plotted against fibre orientation for a specific order or natural frequency, as in Figures 5, 6 and 7. A few selected plots of normal modes, Figures 8 to 11, show the variation with fibre orientation.

Generally, the normal mode shapes corresponding to the first natural frequency do not change significantly with fibre orientation; the variation of tip total deflection of the beam with fibre orientation can be approximated by a straight line, as in Figure 5. The normal mode shapes corresponding to the second natural frequency change quite noticeably for fibre orientations between 1 and 10 degrees. Moreover, at these fibre orientations there is

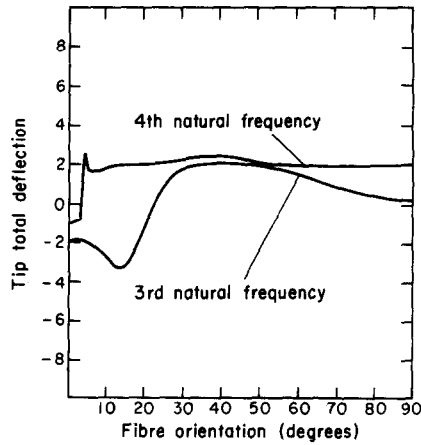


Figure 6. Variation of tip total deflection corresponding to third and fourth natural frequencies of a cantilevered graphite/epoxy beam with fibre orientation.

a change of the rank of vibration from first predominantly torsional mode to second predominantly torsional mode. For fibre orientations greater than 10 degrees, the normal mode shapes do not change significantly. The variation of normal modes with fibre orientation is shown in Figure 8. The variation of tip total deflection with fibre orientation is shown in Figure 5. The curve crosses the zero line between 4 and 5 degrees indicating a change in rank of vibration. The normal mode shapes corresponding to the third natural frequency change quite noticeably between 16 and 24 degrees, and gradual change is observed for other fibre orientations. The variation in normal modes with fibre orientations is shown in Figure 9. Again, a change of rank of vibration, from second predominantly torsional mode to third predominantly torsional mode, is observed. The variation of tip total deflection

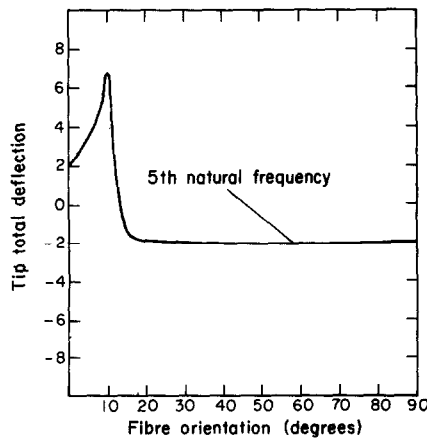


Figure 7. Variation of tip total deflection corresponding to fifth natural frequencies of a cantilevered graphite/epoxy beam with fibre orientation.

with fibre orientation is shown in Figure 6. The tip total deflections of beams corresponding to the fourth and fifth natural frequencies are shown in Figures 6 and 7, and Figures 10 and 11 show the variation of normal modes with fibre orientation.

Discontinuities are observed in tip deflection curves corresponding to the second and fourth natural frequencies, and are confined to fibre orientations less than 20 degrees, where the torsion-flexure coupling effect is relatively stronger. Discontinuity is caused by the fact that the rate of change of frequency leading to a change in mode of vibration differs

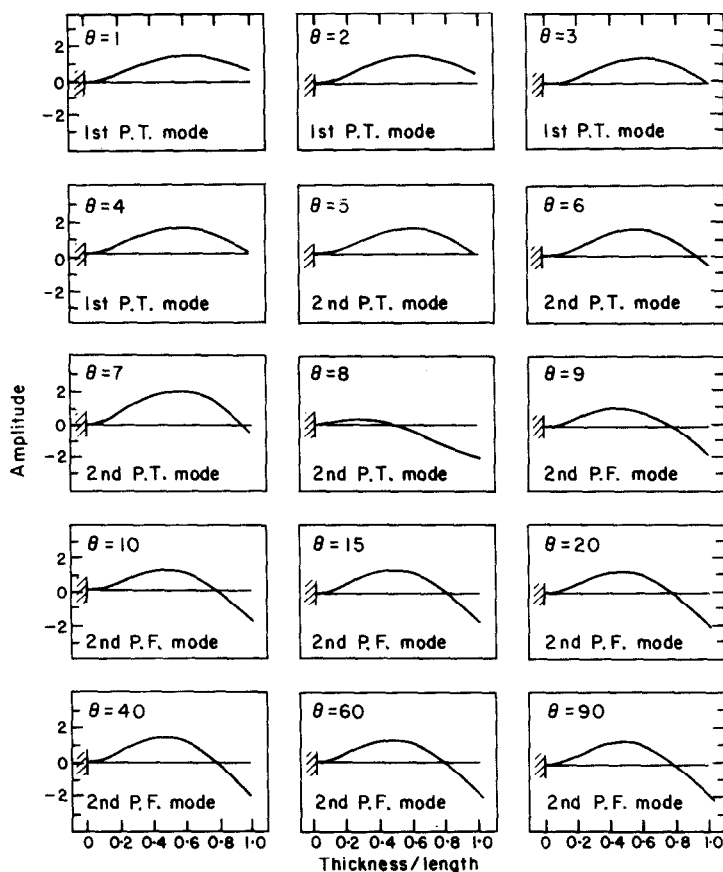


Figure 8. Mode shapes corresponding to the second natural frequency of cantilevered graphite/epoxy beams. θ , fibre orientation in degrees; P.T., predominantly torsional; P.F., predominantly flexural.

as fibre orientation varies continuously. Thus, for instance, the frequency of the second predominantly torsional mode is less than that of the second predominantly flexural mode at a fibre orientation of 6 degrees, while at a fibre orientation of 10 degrees the situation is reversed, hence changing the nature of the normal mode shapes corresponding to these fibre orientations.

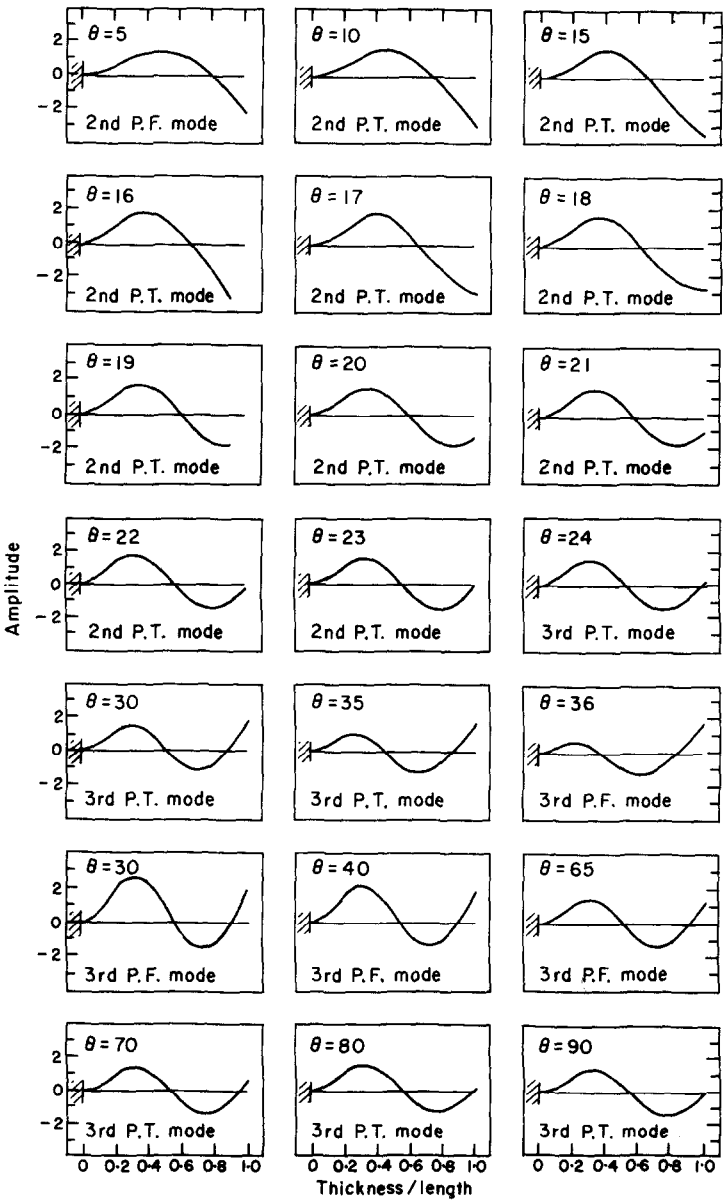


Figure 9. Mode shapes corresponding to the third natural frequency of cantilevered graphite/epoxy beams. Key as Figure 8.

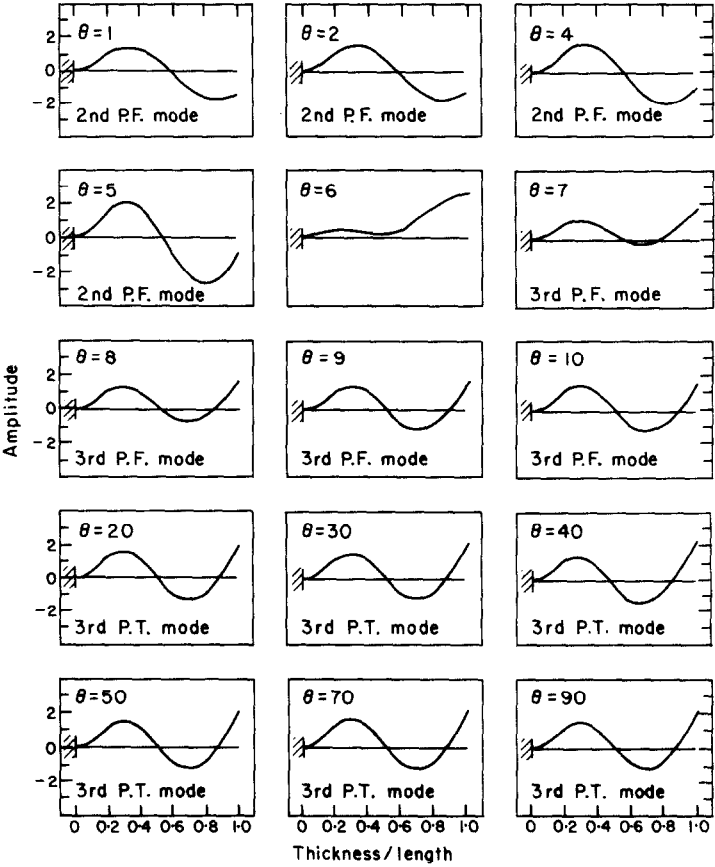


Figure 10. Mode shapes corresponding to the fourth natural frequency of cantilevered graphite/epoxy beams. Key as Figure 8.

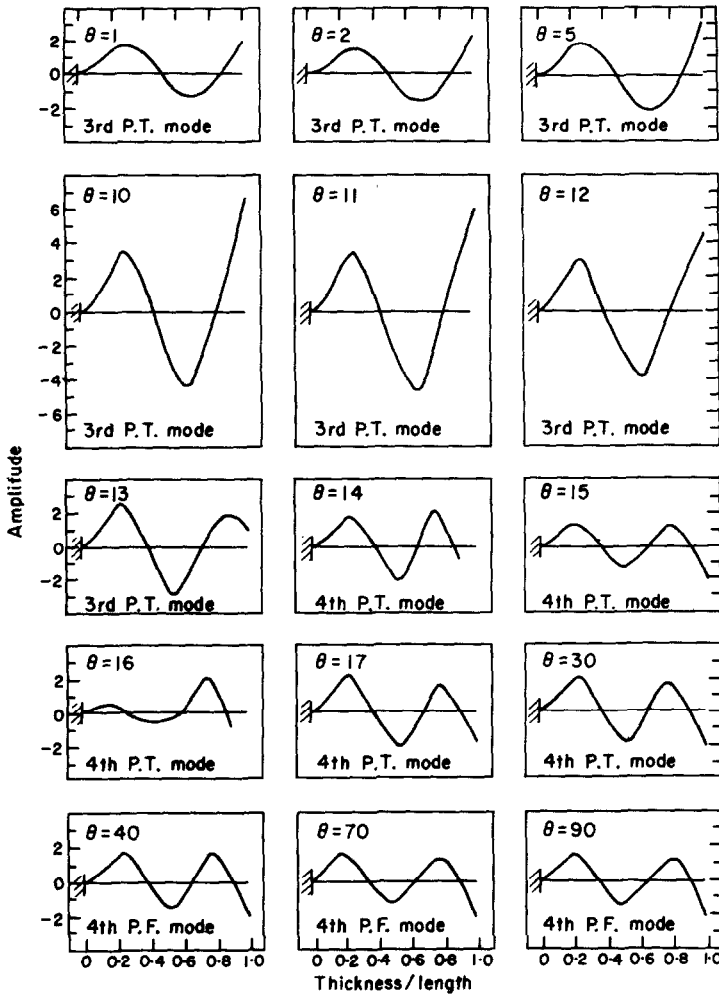


Figure 11. Mode shapes corresponding to the fifth natural frequency of cantilevered graphite/epoxy beams. Key as Figure 8.

4. CONCLUSION

The torsion-flexure coupling effect is due to bending induced by twisting and twisting induced by bending. The present study reveals that these two coupling effects vary with fibre orientation; for a similar range of fibre orientations, these coupling effects always exist. At a higher rank of vibration, the torsion-flexure coupling effect is principally contributed by the twisting moment induced by bending, and the coupling effect is most significant for fibre orientations less than 25 degrees. Within this small range of fibre orientations, the vibrational modes of generally orthotropic beams can experience sudden changes with small increases in fibre orientation.

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APPENDIX: NOMENCLATURE

A	beam cross-sectional area
b	non-dimensional natural frequency $(\rho A/E_{zz}I_x)^{1/2} L^2 \omega$
C_{mt}	mutual stiffness coefficient
C_t	torsional rigidity
E_{ii}	Young's modulus ($i = 1, 2, 3, z$)
G_{ij}	shear modulus ($i, j = 1, 2, 3$ or $i, j = x, y, z$)
I_x	moment of inertia about x axis
J_z	polar moment of inertia about z axis
L	length of beam
M	bending moment
q^2	$J_z/(AL^2)$
r^2	$I_x/(AL^2)$
s^2	$(E_{zz}I_x)/(\kappa G_{zy}AL^2)$
e^2	E_{33}/E_{zz}
T	twisting moment
v	total transverse deflection
η	$1 - \{(E_{zz}I_x C_t)/C_{mt}^2\}$
λ	$C_t/(E_{zz}I_x)$
κ	shape factor
ψ	bending slope of beam cross-section
ϕ	angular displacement of beam cross-section
ρ	density